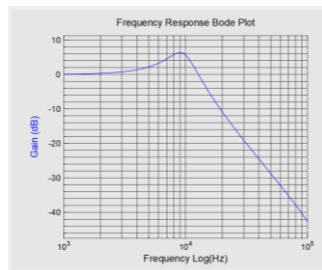


## HW Subsystem Test Plan Example

References: HW\_Testing.docx , Checklist\_Marks.xlsx

| Test item, Description                                                           | Test Conditions                                                     | Expected, Tolerance                                                           | Measured Result                                  | Comparison, Verdict                                  | Remarks, Discussion |
|----------------------------------------------------------------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|--------------------------------------------------|------------------------------------------------------|---------------------|
| <b>Signal Gain-Phase</b><br>- w/o radar sensor<br>- on I- and Q-paths            | Sine input signal:<br>50 uVpp & 1mVpp<br>10 Hz ...                  | Gain in passband:<br>... dB $\pm$ ... dB<br>cut-off frequencies<br>LP/HP: ... | Scope<br><br>Option: Bode Plots<br>(see example) | <b>passed / as designed</b><br><br><b>not passed</b> | ...                 |
| <b>DC level</b><br>- w/o radar sensor<br>- on I- and Q-paths<br>- after each Amp |                                                                     |                                                                               |                                                  |                                                      |                     |
| <b>Radarsignal Gain-Phase</b><br>- on I- and Q-paths                             | Radar Doppler Target<br>K-DT1 with $f_d = \dots$<br>at distance ... | I/Q-Gain-Difference<br>...<br>I/Q-Phase difference<br>$90^\circ \pm \dots$    | Scope Channel A/B                                |                                                      |                     |
| <b>Additional Features</b>                                                       |                                                                     |                                                                               |                                                  |                                                      |                     |

**Example:** Frequency Response Bode Plot (PicoScope 2206B: <https://www.picotech.com/downloads>,  
 3<sup>rd</sup> party WINDOWS app: FRA For PicoScope: <https://bitbucket.org/hexamer/fra4pico/branch/master>)



## FW Subsystem Test Plan Example

References: HW\_Testing.docx, Checklist\_Marks.xlsx

| Test item, Description                                            | Test Conditions                                                            | Expected, Tolerance                                          | Measured Result                    | Comparison, Verdict                              | Remarks, Discussion |
|-------------------------------------------------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------|------------------------------------|--------------------------------------------------|---------------------|
| <b>ADC Samples</b><br>- on I- and Q-paths                         | no signal<br>...                                                           | DC-value of virtual GND<br>...                               |                                    | <b>passed / as designed</b><br><b>not passed</b> | ...                 |
| <b>FFT output</b><br>- on display                                 | Sine input signal:<br>... Hz, ... Vpp                                      | peak and leakage                                             |                                    |                                                  |                     |
| <b>Frequency / Speed</b><br>- w/o sign!<br>- on I- and/or Q-paths | Sine input signal:<br>... Hz, ... Vpp                                      | Frequency $+... \pm \Delta f/2$<br>( $\Delta f = f_s/NFFT$ ) | speed or frequency<br>without sign |                                                  |                     |
| <b>Frequency / Speed</b><br>- with sign                           | sin/cos signals from<br>Matlab and USB-Audio<br>Interface: ... Hz, ... Vpp | Frequency $... \pm \Delta f/2$<br>( $\Delta f = f_s/NFFT$ )  | speed or frequency<br>with sign    |                                                  |                     |
| <b>Additional Features</b>                                        |                                                                            |                                                              |                                    |                                                  |                     |